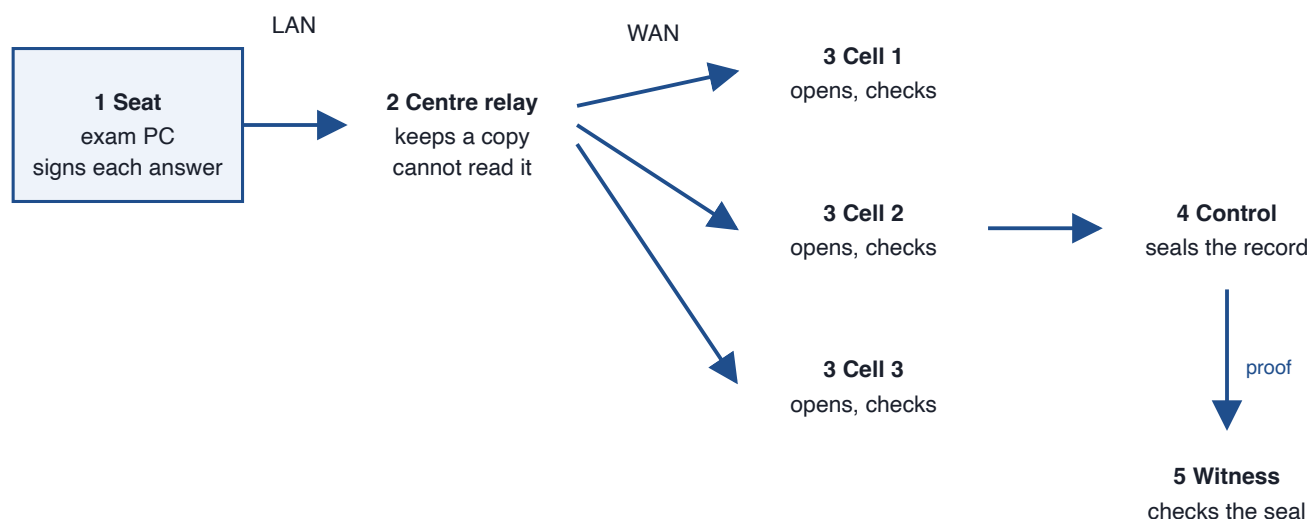


How Saakshi is built

Technology architecture: the parts, the tools we chose, how it stays safe and how it scales.

Big idea: **the seat signs every answer, and nothing in the middle is trusted.** Each part only has to be checked, never believed.

The whole system on one page



Who can see what

Seat: holds the candidate's answers and the signing key.

Relay: sees only locked envelopes. It can store and forward, nothing else.

Cells: 3 independent copies. Any one can be lost and rebuilt from the relays.

Control: signs the final record and runs the audit. It also serves the public check page.

Witness: a separate process that signs only a record that was never rewritten.

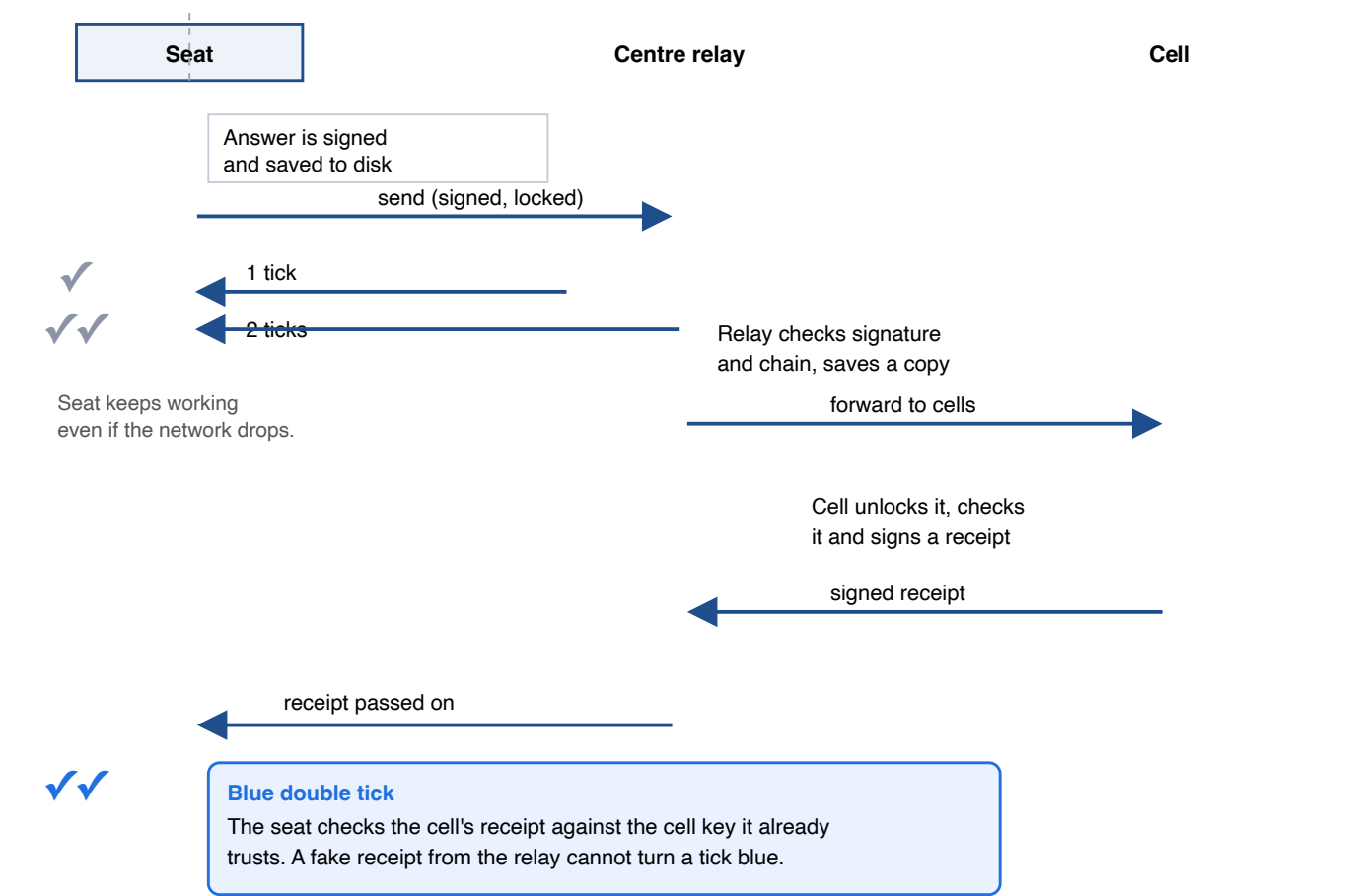
In the demo control and witness share one machine. A separate organisation for the witness comes next.

The data flow, step by step

- 1 The candidate taps an answer. The seat signs it and saves it to disk.
- 2 The seat sends it over the local network to the centre relay.
- 3 The relay checks the signature and the chain, then sends a copy to the cells.
- 4 A cell unlocks the answer, checks it and signs a receipt back.
- 5 Control seals each shift into one fingerprint. The witness checks it.

One answer, three ticks

Like a chat app, the candidate sees how far the answer has travelled. Each tick needs a real check, not a promise.



Tick	What it means	If it fails here
✓	Signed and saved on the seat's own disk	Nothing is lost except the last few seconds. The candidate moves to another seat.
✓✓	The centre relay holds a checked copy	The relay buffers. Answers keep flowing when the link is back.
✓✓ blue	A cell has opened, checked and signed for it	Safe even if the relay or one cell is destroyed.

Technology choices

Part	What we use	Why, in plain words
Exam app on the seat	Electron, React, TypeScript	One app for macOS and Windows. It can run offline and look at running programs on the machine.
Servers (relay, cell, control, witness)	One Bun program with four modes	One thing to ship and test. Each mode is a different job.
Storage	SQLite, with full sync to disk on every write	No database server to run. An answer that was acknowledged survives a power cut.
Signing	ECDSA P-256 and SHA-256	Standard, widely reviewed, built into the platform.
Locking answers	ECDH, HKDF and XChaCha20-Poly1305	Only the cell can open an answer. The relay cannot.
Tamper log	Merkle tree (RFC 9162)	One short fingerprint proves the whole record is unchanged.
Key custody	Shamir secret sharing, 2 of 3	No single person can open the paper before the exam.
Risk analytics	Python with numpy, scipy, networkx	Strong maths tools. Control reads the result back as plain data.
Live views	Server-sent events	Simple one-way live updates for control screens, and they replay what was missed.

Security in five layers

5 Seat gate Blocks remote desktop and overlay tools by name before the exam	What it stops Cheating tools on the PC	
4 Split custody 2 of 3 people must release the keys. A phoned code works offline	One insider opening papers	
3 Independent witness Signs only a record that was never rewritten	A quiet rewrite of history	
2 Merkle log One fingerprint for the whole shift. Any edit is found and located	Edits to the database	
1 Signing Every answer is signed and chained at the seat	Answers changed after the fact	
1,000 / 1,000 tampered records found and located in the tamper lab	0 answers lost in 20 chaos runs	0 honest candidates flagged, and no one is auto-penalised

Honest limit: a signature proves the record was not changed after the device made it. It does not prove who chose the answer.

Scalability: what we measured

A real load test on one laptop. 99 centres and about 20,000 candidates sent answers to 3 real cells.

19,803 candidates unlocked, of 19,803	227,882 answers stored, 0 rejected	44 ms median time to reach a cell (99th: 168 ms)
1.06 MB centre to cell traffic per candidate hour	2.3% seat CPU, median (max 4.3%)	166 MiB seat memory

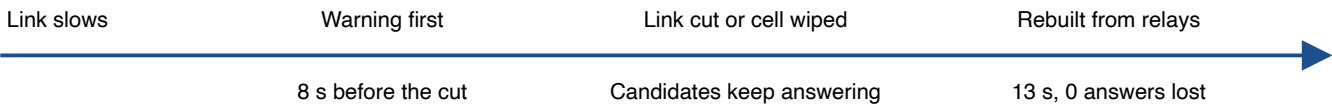
Answers are spread over 3 cells, nearly evenly

Cell 1: 77,338 answers	Add centres: add relays. Each centre has its own.
Cell 2: 75,992 answers	Add load: add cells. Any cell can be rebuilt.
Cell 3: 74,552 answers	Analytics run on an export, off the exam path.

Measured on one Apple M5 Mac with 16 GiB, on the local network, with synthetic candidates. This says nothing about real wide-area links. The per-hour traffic figure stretches a 60 second window, so it is not a real exam duty cycle.

Staying up when things break

20 / 20 chaos runs passed (cell killed, cell wiped, spare relay)	0 of 16,000 answers lost	1 to 1.2 s median time to recover
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Tested on macOS and Windows.

Repository: <https://github.com/anirudh12032008/saakshi>